



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: CS3251 & Programming in C

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Functions and Array of Pointers

- **Course Outcome:** CO3

- **Topic Learning Outcome:** TLO7 & TLO8

- **Activity Chosen:** Debugging

- **Justification:**

Debugging is a crucial aspect of innovative practices in software development and other fields. It involves identifying and resolving errors or issues in a system to ensure it functions correctly and meets its intended purpose. Students can implement the necessary changes to resolve the bug, ensuring the fix doesn't introduce new issues. Debugging is not just about fixing errors; it's about understanding the skeleton and concept of the program.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- The time allotted for the topic is 15 Minutes.
- Different program coding was displayed on the screen with errors.
- Based on the topics discussed about functions, pointers and array of pointers, students started debugging the code.
- Students identified errors on the programs shown on the screen and written the correct code in their notebook along with output.
- Students were able to recognize the type of error and corrected it. Along with debugging code modification without changing the logic was also done for few small programs.
- Students those who have completed the programs without any errors were appreciated.
- The students completed the activity and the sample of activity is shown in Figure 1 and 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	2	1	1	1	1	1	1	1	3	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	1	1	1	1	1	3	1	1
Justification for Correlation	The students will apply the basics mathematics , science and engineering principles for solving problems using modular programming approach.	Students will be able to identify various issues in static memory allocation and provide solutions using dynamic memory management (pointers)	Students will be able to provide solution to the problems by using function concepts and implement in a modular way.	Students will be able to choose the proper functions and pointers to interpret the data to provide valid conclusion.	Students will be able to use DEV C++ software / Linux to provide solution to the problems.	The students will adhere to the ethics.	The students will work effectively as an individual and team for solving the problems using functions and pointers.	The students will be able to expose their knowledge of solving the problems by giving presentation and through discussion forum	Students will improve their knowledge in C continuously by implementing in emerging technologies.	The students will be able to use the concepts of functions and pointers in developing solution for complex computational problems	The students will use the modular programming concepts for designing IoT system and Mobile application	The students will demonstrate modular programming concepts in developing software projects.

- Images / Screenshot of the practice:

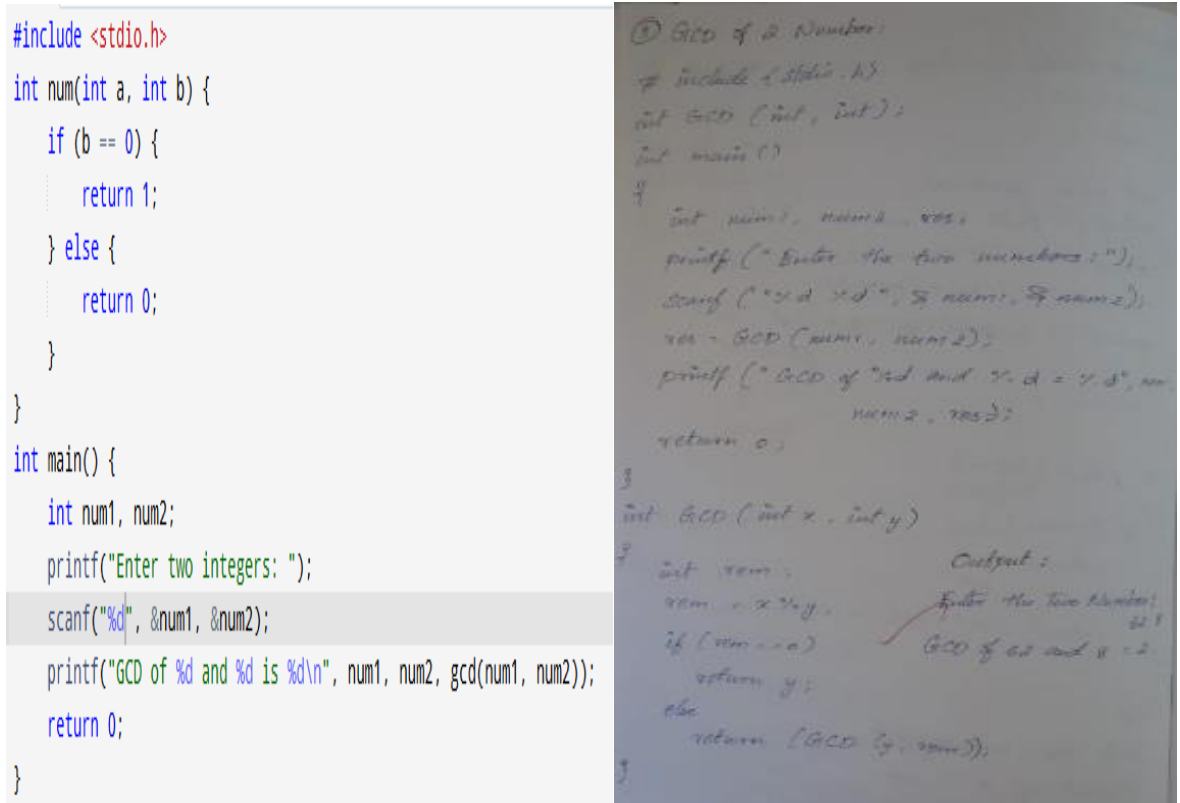


Figure 1. a Debugged code sample of G.Raagul

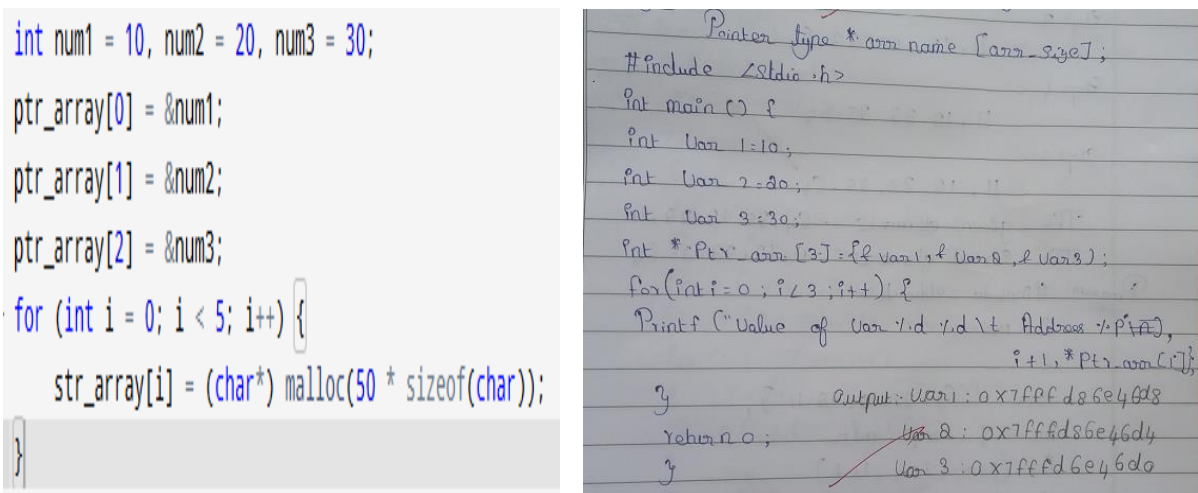


Figure 1. b Debugged and replaced malloc in the given code sample done by S.Saranya

Reflective Critique:**❖ *Feedback of practice from students and other stakeholders:***

- Some of the students completed eagerly and shown a good response.
- Some of them felt difficult to identify the exact logical and conceptual errors, but they were able to find the syntax errors.

❖ *Benefit of the practice:*

- Students treated debugging as an opportunity to learn from mistakes and improve coding practices.
- Students shared debugging experiences and knowledge with their peer students and enhanced team skills.
- This practice improves the analytical skills of the students.

❖ *Challenges faced in implementation:*

- Few students were lagging in identifying the errors.
- Few of them mentioned that identifying and writing consumes more time.
- After the final discussion, they acquired pretty much clarity than before.

References:

- <https://wearebrain.com/blog/10-effective-debugging-techniques-for-developers/>
- <https://dev.to/surajondev/debugging-techniques-every-developer-should-know-85f>
- https://digitalcommons.usu.edu/eled_support_pubs/5/
- <https://niveussolutions.com/debugging-techniques-best-practices/>

Signature of Faculty Member**HOD**